

Digital Food Report: Hardware and Design

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1 Introduction

The Digital Food Team at the Creative Machines Lab mainly works to actively develop a 3-D food printer. The idea behind the project is to re-appropriate the mechanisms used in standard additive manufacturing (FDM most notably) and build a functional printer that extrudes food instead of thermoplastics.

The project works three-fold with software, mechanical, and creative/design implementation. The print starts with a CAD design, each part file counting as a different ingredient. That CAD is exported as an STL and sent to the slicing software (which was coded using the baseline open-source slicer software, Slic3r, for the project's specific design needs). Print characteristics like extrusion rate and tool post assignment are inputted. The Slicing software inputs all the information and compiles a GCode that can be remotely uploaded to the printing apparatus.

2 Goals and Motivation

Work done in previous semesters already confirmed this project works for small prints of ≤ 8 ingredients. The main focus for the Spring 2023 semester was to expand on the fundamental prints already completed to successfully execute an 18-ingredient print. Additionally, the original printer is an older model, so to improve bugs in the electrical and mechanical components of the print, the secondary goal of the semester was to assemble a fully working version of the printer.

3 18-Ingredient Print Design Process

The print design process commenced by drawing from various culinary elements to fabricate a 3-course meal. It was ideal for a large ingredient dish because it separates the 18-ingredient print into smaller assemblies that are less likely to fail.

3.1 Ideation

In this stage, flavor profile, printer compatibility, and the structural integrity of each ingredient were considered when choosing each dish. The dish sought to resemble the prior art with a few dietary shifts. Brainstorming for the menu was detailed as shown in Figure 1.

The final menu was decided based on the aforementioned features and the potential to use ingredients to curate a gourmet palette. The dishes consisted of an egg-less tofu-based quiche, ratatouille held in a structural polenta cup, and tiramisu for dessert.

3.2 Ingredient Testing

The dishes required fresh ingredients that needed processing to a state for facilitated nozzle extrusion. Figure 2 shows the final preparation of ingredients. The list of ingredients included: savory cracker paste, plain tofu, tomato tofu mix, mushroom tofu mix, spinach tofu mix, Ricotta, Mushroom paste, Spinach paste, olive tomato, Olive paste, olive garlic, tomato paste, polenta, squash paste, zucchini paste, tomato paste, ladyfinger coffee paste, cream cheese frosting, and mascarpone. The thicker elements required a significant amount

Figure 1

	Appetizer		Main Course (flavor profile: savory)		Dessert (flavor profile: sweet)	
	dish	ingredients	dish	ingredients	dish	ingredients
isom	quiches (4-ing)	drizzle(filling), egg middle (yellow - structural), pie crust (tan structural), spinach(pepper (red or green structural),	steak fillets (8-ing)	drizzle(filling), lettuce 1 (green-structural), fries (yellow - structural), steak crust (brown structural), steak inside (brown structural), tomatoes (red filling)(structural), onions (purple or white structural), lettuce 2 (green-structural)	cheesecake (6-ing)	drizzle(filling), creamcheese (white - structural), pie crust (tan structural), fruit 1 (red structural), fruit 2(red filling)(structural), fruit 3 (blue structural)
		co-centric circle disks		circles(dark red,brown), lines, spirals for lettuce		cylinder, drizzle
try	salad (6-ing)	lets cheese puree (base, structural), tomatoe (filling), olive puree (filling), cucumber puree (filling), pepper puree (filling), dressing (drizzle)	polenta (5-ing)	polenta (structural), pesto (cup, filling), tomatoe puree (cup, filling), cheese (filling puree), balsamic glaze (drizzle)	tramisu inspired layer cake (7-ing)	cookie crust (base layer, structural), chocolate cream (filling), coffee cream(filling), cookie layer 2 (structural), vanilla cream (filling), chocolate (drizzle)
		circular base with fillings		circular polenta base with cups of filling		cylinder, drizzle
in	Potato Leek Soup	Potato Puree Chicken Stock Purvenzed Leeks Unsalted butter Heavy Cream Salt/Pepper	Vegan Tian (Ratatouille)	Eggplant Puree Roma Tomatoe Puree Yellow Squash Puree Zucchini Puree Bell Pepper Puree Minced Onion Minced garlic Olive Oil (drizzle)	Dark Chocolate Tart	Cookie Crumbs Melted Butter Heavy Cream Dark Chocolate (filling) Dulce De Leche Sauce Caramel Drizzle
		Liquid/Printed in dish with sides		Circular base, layered vegetable filling		Pie shape base with filling
ushka	guacamole	crushed chips (structure), avocado puree (filling), lemon juice (filling), tomato puree (filling), onion puree (filling), srirache (drizzle), salt/pepper (drizzle)	Tostada	crumbled chips (structural), mashed rice (structural), beans (puree), tomato puree, avocado puree, sour cream, melted cheese	Chocolate & Coconut Torte	cookie crust (base layer, structural), Whipped Cream (filling), Whipped Coconut Cream (filling), Melted Chocolate, Coconut Shavings (topping), chocolate drizzle
		printed on flat dish		circular base		cylindrical with drizzle
isa	Pizza bites (8-ing)	dough (structural), tomato paste (structural), cheese (filling puree), ham (filling), chicken (structural), lettuce or tomatoe (green-structural)	Crab cakes (7-ing)	crab (same way we make chicken - structural), ricotta (filling) black bean puree (filling), mango puree (filling), tofu (structural), almond sauce (drizzle), green and tomatoe sauce on top	Lemon curd pie (5-ing)	graham cracker (structural), lemon curd (structural), vanilla frosting (structural), everything else would be any looking circular splitches on the top
		circular dough base, second circular layer of tomato paste, the rest acting as toppings		circular base for crab, ricotta in a smaller concentric circle on top, puree and tofu, decorating the sides, greens and tomatoe sauce on top		circles for graham cracker and lemon, everything else would be any looking circular splitches on the top

Figure 1: The google sheets page detailing the options for dishes in the print

of processing to improve the rheology. All the raw vegetables or blended tofu mixes required a blending component with water and some form of a thickening agent.

Figure 2



quiche1_1_scaled	Savory cracker paste	structural	☒	1
quiche1_2_scaled	Plain tofu	structural/filling	☐	2
quiche1_3_scaled	tomato tofu mix	structural	☐	3
quiche1_4_scaled	Mushroom tofu mix	structural	☒	4
quiche1_5_scaled	spinach tofu mix	structural	☒	5
quiche1_6_scaled	Ricotta	filling	☐	6
quiche1_6_scaled	Mushroom paste	topping	☒	7
quiche1_6_scaled	Spinach paste	filling	☐	8
quiche1_6_scaled	olive tomato	filling	☐	9
quiche1_6_scaled	Olive paste	filling	☐	10
quiche1_6_scaled	olive garlic tomato paste	filling	☐	11
ratatouille1_1_scaled	Polenta	structural	☐	12
ratatouille1_2_scaled	Squash paste	structural	☐	13
ratatouille1_3_scaled	Zucchini pasta	structural	☐	14
ratatouille1_3_scaled	Tomato paste	structural	☐	15
tirimisu1	Ladyfinger coffee paste	structural	☒	16
tirimisu2	cream cheese frosting	structural	☐	17
tirimisu3	Marscapone	structural	☐	18

Figure 2: Third print ingredient list, with syringes prepped for the print

While the recipe detailed for the third print incorporates fresh ingredients, processed ingredients tend to extrude much better for the syringes. With the previous recipes mainly consisting of vegetables, water, and xanthan gum, the vegetable purees had more of a gelatin consistency, which did not extrude as well. The final print's recipe was adapted to incorporate the original ingredients but used a neutral paste as a thickening agent rather than the xanthan gum to improve the rheology. This recipe included tahini since its texture rivals peanut butter, but it is much more compatible with the savory flavor profile than peanut butter. Food coloring enhanced the aesthetics of the final print, especially since many commercially available produce and food products undergo some form of coloring to

enhance the appeal. The final documentation of the recipe is depicted in Figure 3.

Figure 3

	true recipe	food coloring ratios (drops)		final recipe	extrusion rate
quiche1_1_scaled	Savory cracker paste	structural	☒	1 cracker paste	0.028
quiche1_2_scaled	Plain tofu	structural/filling	☐	2 egg, water, cream cheese	0.02
quiche1_3_scaled	tomato tofu mix	rfc	☐	3 egg, tomato, water, cream cheese, food coloring	0.02
quiche1_4_scaled	Mushroom tofu mix	r+gfc (1:1)	☒	4 egg, mushroom, water, cream cheese, food coloring	0.02
quiche1_5_scaled	spinach tofu mix	gfc	☒	5 egg, spinach, water, cream cheese, food coloring	0.02
quiche1_6_scaled	Ricotta		☐	6 ricotta	0.024
quiche1_6_scaled	Mushroom paste	r+gfc (1:1)	☒	7 mushroom, water, tahini, food coloring	0.025
quiche1_6_scaled	Spinach paste	gfc	☐	8 spinach, water, tahini, food coloring	0.025
quiche1_6_scaled	olive tomato	rfc	☐	9 olive, tomato, water, tahini, food coloring	0.025
quiche1_6_scaled	Olive paste	y+r+bfr(4:2:2)	☐	10 olive, water, tahini, food coloring	0.025
quiche1_6_scaled	olive garlic tomato paste	r+gfc(3:3)	☐	11 olive, tomato, mushroom, water, tahini, food coloring	0.025
ratatouille1_1_scaled	Polenta	structural	☐	12 cracker paste	0.028
ratatouille1_2_scaled	Squash paste	yfc	☐	13 Squash, water, cream cheese, food coloring	0.025
ratatouille1_3_scaled	Zucchini pasta	gfc	☐	14 Zucchini, water, tahini, food coloring	0.025
ratatouille1_3_scaled	Tomato paste	rfc	☐	15 tomato, water, tahini, food coloring	0.025
tiramisu1	Ladyfinger coffee paste	structural	☒	16 graham cracker paste	0.028
tiramisu2	cream cheese frosting	structural	☐	17 cream cheese frosting	0.0195
tiramisu3	Marscapone	structural	☐	18 chocolate pudding	0.025

Figure 3: Final ingredient list with food coloring ratios

3.3 CAD Modeling

The CAD designs emulate the aesthetics of the previous dishes, such as the crust of the quiche being a miniature pie-like structural host.

The first sketch, as shown in Figure 4, was used to map out how each ingredient would sit in the assembly. This image maps out a rough categorization of structural and filling materials and how they would come together.

Figure 4



Figure 4: Original brainstorm sketch of the ingredient breakdown for the print

When modeling the CAD, the rough sketch plotted out the ultimate compositions of each sub-assembly. Each part inserted in the assembly correlates to an ingredient. For

example, if multiple copies of the part files are in the CAD assembly, they will each count as a different part file and correspond to a unique ingredient.

Figure 5 shows the initial designs for the ratatouille dish and the quiche. Both designs required a cupped structural material to hold the less stiff material. The cup design is the largest, the figure shows a volume related to each file. Each ingredient should not exceed the maximum amount of material a syringe can hold.

Figure 5

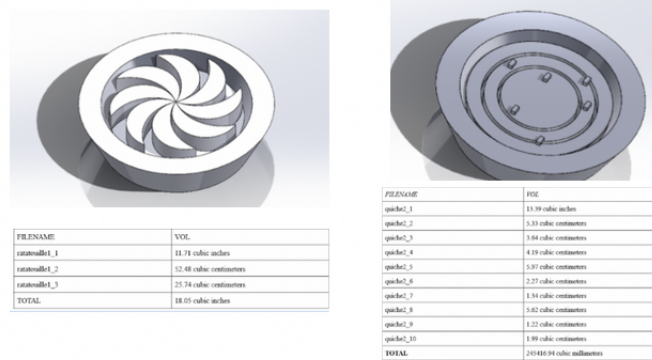


Figure 5: CAD File of the first design of the print in Solidworks

The second rendering of the assembly, shown in Figure 6, includes a design for the tiramisu. It is a three-part hexagonal base that lofts into a circle with a slight rotation. Also, features that favor the print include taking the stylized outward draft on the wall of the quiche and ratatouille cup. Unlike an FDM printer, food can not draft outward or generate supports for complex geometries. Printing food favors tapered inner walls for the cup design and was adapted in the second iteration of the print.

Figure 6

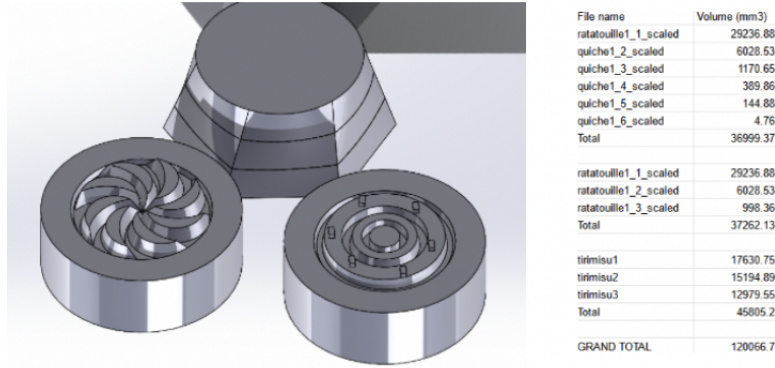


Figure 6: CAD File of the second design of the print in Solidworks

The next iteration of this design, as depicted in Figure 7, reflects a further optimized design where the part files were scaled down. These design changes were necessary after the first print iteration failed due to over-extrusion of the ingredient. The part files needed

scaling down since the failed part file occurred twice. The vertical walls were tapered inward for a stable base, and the volumes were much smaller than provided in the original print.

Figure 7

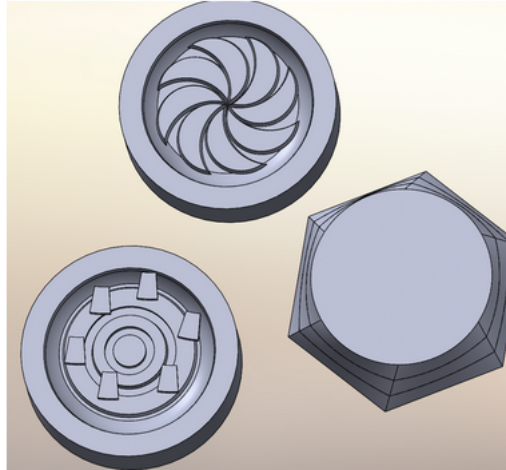


Figure 7: CAD File of the third optimized design of the print in Solidworks

The final iteration of the print included design changes in the ratatouille to make the filling ingredients more visible despite consistency. This design largely kept all other dishes the same, but the ratatouille now has defined petals embedded in the previous layer, so each print has its clear section.

Figure 8

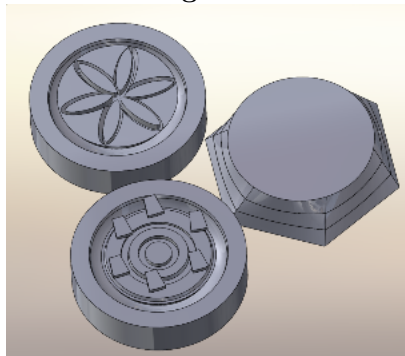


Figure 8: CAD File of the final optimized design of the print in Solidworks

4 Spring 2022 Prints

4.1 Print 1

The first print failed. The extrusion rate for the savory cracker element was estimated using the extrusion rate of graham cracker paste that follows a similar formula. The print

was over-extruded with the extra material on the sides of the cup shown in Figure 9. The next CAD design featured scaling down, and the extrusion rate was re-calibrated to account for the extra material.

Figure 9



Figure 9: Final state of the first print iteration

4.2 Print 2

The second print resulted in two failures. The CAD model shown in Figure 7 shows the quiche had six small ingredients. The GCode did not slice a path large enough to extrude any material causing the failure. When restarted at the point of the polenta cup, the nozzle was too low, so the tip grazed the bottom of the printer bed and broke off seen in Figure 10. To improve the design of the next print the ingredients on the quiche surface were enlarged, and the CAD design in Figure 7 was used in the third print iteration.

4.3 Print 3

The third print was the first print where all ingredients were successfully deposited on the bed, although the polenta was over-extruded.

When the print ended, the Y axis homed in the wrong direction. It caused an array of problems including a misaligned axis and broken brackets holding the gantry in place before the next print.

4.4 Print 4

The fourth print required quite a bit of maintenance on the machine before beginning. Numerous mechanical failures occurred within the first few attempts. Some fixes included

Figure 10

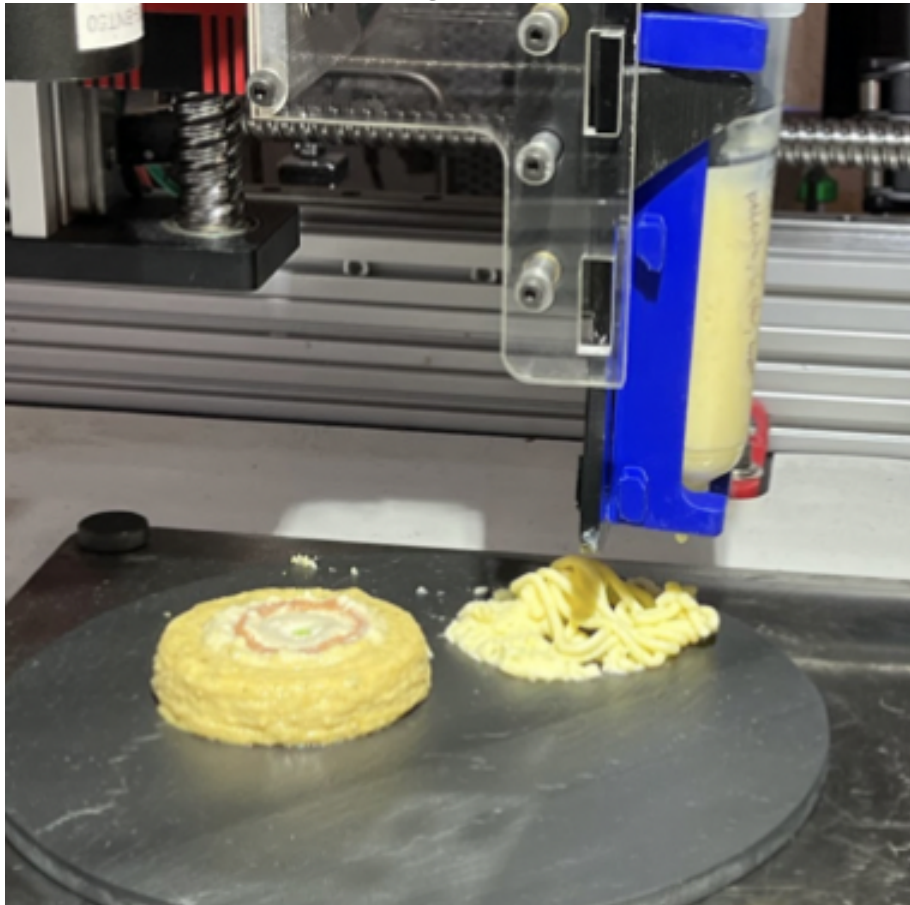


Figure 10: Final state of the second print iteration

re-securing the L brackets, replacing u-extruder parts, and completely realigning the x and y-axes. The first two attempts on the final print failed due to the left y motor switching polarity despite the coordinates fed on the hatching pattern, breaking a few components in its path. Y-axis failures became a consistent theme with this print indiscriminate manual instructions or the G-CODE parsing.

The first attempt at resolving the y-axis issues was to make changes in the G-CODE to slow the printer down in the hatching patterns since most failures occurred there. The conjecture was that the printer had errors receiving the information at the default speed, so the infills printed at a reduced feed rate. The manually adjusted G-Code had speeds maxing out at 800 since the rates were reduced from F1200 to F800 and F900 to F600. The new G-Codes did not help the situation much as the print failed on the first layer on the 3rd time and the third layer on the 4th also as seen in Figure 12. The newest conjecture is that an unknown trigger is hit at these points in the print because the machine also homed the y-axis in the opposite direction of its program when instructions were manually fed from the printing UI after the failure. Before the 18-ingredient print can be attempted again, the

Figure 11



Figure 11: Final state of the third print iteration

issue (potentially wiring or software related) causing the y-axis to sporadically flip polarity must be diagnosed and amended.

4.5 Print 5

After letting the printer rest, the final 18-ingredient print was successful shown in Figure 13. Using the same G-Code and ingredient composition as the precedent helps indicate the previous failures were a result of overuse. While it was proved the machine can execute the 18-ingredient print, it needs a ton of work done to be reliable.

5 Printer 2.0 Development

In addition to the food prints, the Food Printer 2.0 machine has been updated to make it operational for future prints. The main difference between this new machine and the previous iteration is its onboard controller. The new gantry incorporates a Duet 3 Mainboard 6XD as its primary controller board, whereas the current gantry uses a Duet 2 board. Table 5.1 lists some of the spec differences between the two boards.

Table 5.1: Duet Board Specifications

Figure 12



Figure 12: Y-Axis polarity failure on fourth print iteration

	Duet 2	Duet3 Mainboard 6XD
Processor Speed	120 MHz	300 MHz
IO Connections	5	9
Firmware	RepRapFirmware 2.x	RepRapFirmware 3.x

Upgrading the controller board will hopefully mitigate some failure modes that have caused several prints to fail. As discussed previously, the dual Y-axis motors will intermittently become out-of-sync with each other, causing the gantry to come off its axis and break the printer alignment. The suspected cause of this failure is the Duet 2 processor sending out motor commands too slowly, causing the de-sync. The increased processing speed of the Duet 3 Mainboard should prevent this failure from re-occurring on the new gantry.

The two major tasks in getting the printer built and operational were building the U-axis for the extruder motor and setting up the software interface and homing methods. The U-axis for the new printer operates the same way as the old printer. Table 5.2 lists the ordered parts, their cost, and the vendor for the new axis.

Table 5.2: Food Printer 2.0 Parts List

Part	Vendor	Cost
GT Timing Belt Mount (24 teeth)	McMaster-Carr	\$13.17
GT Timing Belt, 6mm wide x3	McMaster-Carr	\$22.86
Lead Screw, 1/4", 3 ft	McMaster-Carr	\$4.74
Bronze Sleeve Bearing	McMaster-Carr	\$2.67
	Total Cost	\$43.44

Figure 13



Figure 13: Final state of the 18-ingredient print

After constructing the U-axis, the limit switches were mounted at the home motor positions for all four axes. The Duet interface was then configured, and the printer was successfully homed. With the software and hardware setup complete, the printer is now operational.

6 Future Work

Food Printing is an ever-expanding field of study, with new ingredients and printing techniques open for discovery. Now that the 18-ingredient milestone was attained, future work for the digital food printing team will focus on expanding the menu of dishes. Much of the difficulty with constructing a meal through additive manufacturing is manipulating the rheology of raw ingredients into shear-thinning pastes easily extruded while still holding their shape after deposition. By expanding on the database of available recipes, we can create new and innovative flavor profiles to experiment with in our prints.

The digital cooking field develops alongside our prints. In previous trials, a blue-light laser (connected to the gantry) broiled the dishes during prints. Some other heating apparatus' to be experimented with include IR spot heaters, which emit a focused beam of IR radiation onto the cook surface, and halogen bulbs with reflective heat shielding to direct heat onto the print bed. By incorporating these new heating technologies, we will be incorporating a crucial element to the technology.

7 Summary

The Digital Food Team at the Creative Machines Lab has developed an 18-ingredient food print in the form of a 3-course meal consisting of quiche, ratatouille, and tiramisu. The CAD model of the 18 ingredients was crafted and exported as STL files in Solidworks. After many changes, two successful iterations of the print were completed by the end of the semester. There were over five attempts at the print, where common failures led to orientation issues with the printer hardware and necessitated slight changes in CAD, extrusion rate, and videography methods. Finally, with the hardware and software setup of the Food Printer 2.0 complete, future prints will be executed without many of the issues seen with the current printer.